Chapter 16

REGIONAL POLICIES IN DEVELOPING COUNTRIES

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1. Introduction

There are about one hundred countries in the world which may be regarded as "developing", excluding the small island states and principalities with populations of under one million. This total encompasses a huge range of geographical and economic diversity, from the high-income Middle Eastern oil exporting countries, to the prosperous city states of Hong Kong and Singapore, the emerging middle-income South American and Mediterranean countries, and the low-income high-density Asian countries. This diversity is compounded by marked differences in government structures, in social objectives, and in cultural values and traditions. Some countries have high levels of urbanization with only low levels of industrialization. Others have very centralized urban hierarchies and spatially concentrated economies. Yet others have a spatial distribution which reflects strong tribal divisions or ethnic diversity. All commentary on economic development issues runs the risk of facile generalization, because there always appear to be exceptions to any attempt to discern a pattern.

Diversity in the regional problems of developing countries and in the social and economic policies designed to alleviate these problems reflects not only different levels of development and systems of government but also geography, history and tradition, and societal goals. In any country, developed or developing, what constitutes a regional "problem" differs among social groups. In the developed industrial market economies, regional problems are rarely defined in terms of absolutes. *Relative* imbalances in indicators of the social and economic conditions of different regions induce regional policy responses from the centre. A regional problem can be defined only in the context of the country's dominant social and political values. In many developing countries, on the other hand, the interregional imbalance factor is often given a low priority in the portfolio of national economic and social policies. Development and economic growth in the national setting are much more likely to be the dominant policy objectives, with the spectre of rapid population growth often lurking near the foreground.

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In the developing country context, a distinction has to be made between regional policies which are primarily a response to political pressures and regional policies which are essentially regionally based or focused on economic development programmes, often within the framework of a national economic development strategy. Political pressures for differentials in public expenditure favouring a given region or group of regions may reflect the supporters of the ruling party or elite and their geographical base. The central government may also need to smooth tribal, ethnic or religious divisions with regional aid programmes. A sensitive frontier region may also be favoured, not only in logistically useful infrastructure investments (e.g. military roads and airbases) but also in the provision of government services. In federal states, the revenue-raising powers of individual states may be reinforced or distorted with differential federal government grants and subsidies. Non-federal states, which tend to have more primate urban hierarchies [Henderson (1982)], may favour expenditures in the largest city or cities; or may favour the smaller cities if the primate city is perceived to be "too big". Although political pressure is exercised on behalf of both *places* and social *groups*, regional economic development policies tend to be place-specific, focusing only indirectly on target groups.

The economic case for regional economic policies in developing countries follows the well-established case made out for such policies in developed countries. The case rests partly on the development and utilization of economic resources, including human resources. It also relies on the imperfect operation of factor markets and the tendency for levels of economic prosperity between regions to widen as national per capita incomes rise [before subsequently converging again, Williamson (1965)]. Policy intervention may be necessary to keep the disparities in check. Intervention may also be justified on the grounds that social costs (benefits) are different from private costs (benefits) for both households and firms as a result of rapid urban growth and strong interregional factor flows. Within regions targeted for development, increasing returns to scale implies selectivity in growth (by sector, urban place, or expenditure programme).

The economic case for regional economic policies is also supported by both the "top-down" and the "bottom-up" schools of development planning. On the one hand, the "region" is a useful bureaucratic layer for the application of national policies which require factory or farm gate or doorstep delivery; on the other, it is a vulnerable economic, social and political entity buffeted by strong external forces. For top-down planners, the region is an administrative and managerial convenience, not as subject to local pressures and influence as a municipality but insulated from the centre and day-to-day political control. For bottom-up advocates, the region has to be designed as a political entity with considerable autonomy, seeking the opportunities while avoiding the dangers of closer contacts with the national and the world economic system. Simplifying greatly, both sides view regionally based economic development policies as an effective mecha-

nism towards more rapid overall development, although each defines development in a different way.

One key feature dominates the less-developed nations of the world in contrast to the smaller group of industrialized nations, apart from differences in levels of Gross National Product per capita. This is relatively high rates of population growth, and even faster growth of urban populations. Between 1970 and 1981 [World Bank (1983)], the average annual percentage growth rate of urban population in thirty-four low income nations (GNP per capita of less than 400 U.S. dollars in 1981) was 4.4 (and 5.3 for nations in this group other than India and China). In thirty-nine lower middle income nations (up to $1,700 per capita), the growth rate was 4.3 percent. In twenty-one upper middle income nations (up to $5,500 per capita), the rate was 3.8 percent, and in the four large Middle Eastern oil exporting states, the rate was 8.2 percent. In contrast, in the industrial market economies the rate averaged 1.4 percent, and in Eastern Europe the rate was 1.8 percent. Developing country population growth rates are currently much higher than the growth rates experienced by the now developed countries in their early industrialization phase.

Any attempt to use the experience of developed countries as a springboard for discussion of regional policy problems in developing countries is subject to major constraints. First, as already pointed out, the demographics are so different. Regional policies in countries with rapidly growing populations are focused more on problems associated with the absorption of people in the markets for labour, housing and public services. In countries with stagnant populations, on the other hand, many regional problems are concerned with the effects of depopulation on local economies. The "typical" problem in a developing country is how to generate economic development (and jobs) in a hitherto underdeveloped region relying on traditional agriculture supported by a hierarchy of urban service centres; in developed countries the typical problem is how to revitalize a chronically depressed old-established industrial region (job creation remains important because the rate of job loss often exceeds the rate of population decline).

Second, the environment in which interregional economic change takes place varies so much between developed and developing countries. In many developed countries economic activities could relocate with minor efficiency losses. Even lagging regions have growth potential (many of them grew rapidly in the past) because they have skilled labour forces, infrastructure, agglomeration economies in their cities, and entrepreneurship. Also, they are well connected to the rest of the national space economy via comprehensive interurban transport and communication networks. Moreover, many industries in developed economies are footloose and can operate at a large number of locations at similar cost and with more or less equal access to national markets. If firms remain reluctant to relocate to peripheral regions, the reasons may be more non-economic than

economic. In developing countries, on the other hand, comparative costs may vary widely among locations, access to national markets is often very low outside the core region, and the supply of infrastructure may be very inadequate. Thus, the economic costs of a peripheral location may be too high to be bridged by affordable location subsidies.

A third difference between developed and developing countries is in regional policy objectives and priorities. In developing countries there may be substantial emphasis on promoting economic development and on stabilizing rural regions. In developed countries interregional equity objectives may be dominant, and regional policies may be used to pursue goals such as environmental protection, resource conservation and the control of inflation. The goals adopted in developing countries frequently result in active intervention in rural development strategies by the private sector. The industrial decentralization policies that may exist on paper are only rarely pursued vigorously. The trade-off between growth and spatial distribution goals may be difficult to reconcile. In developed countries the policy goals are much more likely to be compatible. For example, measures to reduce interregional income disparities may have side benefits in moderating the effects of external diseconomies in the prosperous regions and in alleviating inflation [Higgins (1973)].

Of course, in many developing countries spatial equity considerations are given a high priority among regional policy objectives. Examples include Brazil where for more than two decades the Federal government has consistently attempted to promote economic development in the much poorer North East region, and the Philippines where each region has a regional executive office under the National Economic Development Authority (NEDA) and an Integrated Area Development Plan focusing especially on the rural areas. Interestingly, these two cases illustrate a more general hypothesis that countries which emphasize interregional equity tend to have inequitable income distributions.

This raises the possibility that the priority given to regional development policies in some countries may reflect an attempt to substitute spatial equity for interpersonal equity. On this view, regional policies are a smoke screen to divert attention from the economic and social problems that result from the heavily skewed distribution of income.

Section 2 of this chapter discusses the key themes which link thinking about regional development to the process of economic growth. Section 3 refers to the policy implications of these theories. Section 4 argues for the relevance of the urban dimension in regional policy analysis. In Section 5 the discussion turns to four major aspects of regional development strategy: experience with national and regional planning agencies; the role of infrastructure; policies to promote industrial development in lagging regions; and the roles of communications and human resource investments. Section 6 draws some lessons from regional development experience in developing countries in the past two decades. Finally, some

conclusions on the effectiveness of past policies and implications for the future are presented in Section 7.

2. Alternative theories of regional development and economic growth

Several countries have adopted regional development strategies in *all* regions as a policy base for national social and economic development. These countries include Chile, Peru, Venezuela, the Philippines, Tanzania and Zambia. In federal countries, however, the constitutions often give much of the responsibility for subnational development planning to state ministries and agencies. In some cases, a more piecemeal regional approach is adopted, emphasizing the planning of metropolitan regions (e.g. Bombay, Lagos, São Paulo), or resource-rich areas (e.g. the Nabouk Valley in Malaysia, Ciudad Guayana in Venezuela), or river basins (e.g. the Lerma in Mexico, the Euphrates in Syria).

Much of the thrust for regional development strategies and policies in all of these approaches has derived from the need for a multisectoral approach to development, coordinating the skills of architects, engineers, planners and economists and the activities of sectoral agencies at both the national and local levels. In addition, links have been developed between the administrative hierarchies of each country and the structure of political power and control. The specific solutions adopted, however, are different from country to country. Behind all the country differences, an almost universal perception is of a lack of regional "balance" and of spatial distortions in the pattern of national economic growth. In most countries, a regional policy rests on much more than an administrative game plan for more efficient economic growth.

2.1. Spatial disparities

The idea of spatial "distortion" stems from the observation that the geography of economic growth in all nations is uneven. For some, this in itself is undesirable. Over time, the *relative* prosperity of the inhabitants of some localities falls behind, apparently for reasons beyond their control. Political demands develop for corrective action. In particular, "excessive" primacy of urban growth in the largest city or cities is identified as resulting in both economic inefficiencies and social inequities. Lack of "balance" results in strong and polarized migration flows and the possibility of social turmoil.

This dissatisfaction with uneven development may be sufficient grounds politically to identify a regional "problem" and to introduce regional development policies and/or urban development strategies, directed at changing the spatial distribution of population and economic activity. However, in the absence of

explicit social goals and in the absence of analysis as to why the pattern of development is uneven, this can hardly constitute a rigorous case for policy intervention. It is very difficult to argue analytically that social welfare will be unambiguously improved by one pattern of population distribution rather than another, given all the relevant and varied exogenous parameters operating at a specific point in time in a given country.

What analysis in this field has been able to do is to offer reasons why development is likely to be uneven, pointing in particular to three factors: increasing returns to scale, constraints on the free operation of factor markets, and the role of government in an industrializing economy. The economic consequences of these three factors provide economic grounds for policy intervention. It is important to note that analysis in this area has not always observed the distinction between the positive and the normative. The contribution of regional economics to diagnosis has at times become mixed up with the policy guidance of urban and regional planners and policy makers.

The initial prosperity of the majority of third world countries was based on the export of minerals, raw materials and foodstuffs, an orientation frequently associated with colonialism. Economic growth was concentrated on a relatively narrow range of products and was restricted to a small number of urban centres. These centres grew, often as ports or transport interchanges, and a pattern of spatial imbalance developed. Urban primacy and regional concentration resulted from international trade. Profits were derived from natural resource locations and from agricultural areas, and remitted overseas or used to build up capital in the cities. Spatial concentration followed from the distribution of income and wealth and from increasing returns to early infrastructure investments (e.g. railways). Early stages of industrialization followed this pattern, with urban and foreign capital often being used to undercut indigenous producers of simple household goods and farming equipment.

This fact of initial concentration is reflected in the theory of the export base and local multiplier impacts. However, this perspective is too limited once a nation is launched on a path of industrialization and commercialization of agriculture, particularly against the background of continuing high birth rates and falling death rates. Three alternative economic viewpoints may then be taken as to why a spatially uneven pattern of development persists and gets more marked, and may eventually result in a slowdown in the rate of economic growth and a deceleration in the growth of aggregate social welfare.

2.2. *Cumulative causation and core–periphery models*

The first viewpoint began to be defined by Myrdal and Hirschman, and was given prominence by Friedmann in his studies of Venezuela [Friedmann (1966)]. In

Myrdal's analysis a region begins to develop with a "growth trigger" such as a mineral discovery or development of a new export food crop. Rising real wages and high returns to capital reflect increasing returns to scale and the development of spatial external economies of agglomeration, manifested in rising labour and capital productivity as a function of the growth rate of regional output [Kaldor (1970), Oates et al. (1971), Richardson (1973), Dixon and Thirlwall (1975)]. Limits to the increase in the growth rate may be explained by inadequate capacity or external diseconomies. Regional economic growth is, in effect, a process of cumulative causation, in which the high value cards are dealt to those regions which already hold a favoured hand. Any allocation of public expenditure among regions which seeks to maximize returns to the growth of the national product will tend to reinforce regional disparities. Market forces and state spending join in increasing rather than decreasing the inequalities among regions.

Both Myrdal and Hirschman influenced thinking about processes of economic growth among countries as well as among regions within a single country. These ideas were also linked to those of Perroux (1955) whose Schumpeterian theories of innovation were given a geographical focus by Boudeville (1966) and others to explain the processes by which groups of interrelated industries in individual cities or subregions took off onto a path of self-sustaining relatively rapid growth. The literature on growth poles and growth centres is now voluminous. This theory of unbalanced growth has been widely used for policy prescription.

Friedmann's view of the generation of regional inequalities extended the range of relevant factors away from the narrowly economic. His core–periphery model (1966, 1972–73, 1973) views lagging regions as a periphery, standing in a colonial relationship to the core, dependent and lacking economic autonomy. Ideas, technology, capital, attitudes favourable to economic development: all are generated in the core, or received in the core from overseas. Spatial dualism emerges and is reinforced over time. The model incorporates imbalances in political and economic power, and asymmetry among regions in rates of cultural change. These forces influence patterns of innovation diffusion, investment, the allocation of resources by the state and migration. Eventually, however, the disparities among regions begin to narrow. The core–periphery dichotomy becomes blurred as markets expand, communications improve, attitudes change, and urban growth disperses. Key issues are whether this turnaround will ever occur without active policy intervention, and if so whether it makes good sense to try to accelerate the process.

2.3. Radical interpretations

The notion of regional dependency at the heart of the core–periphery model has led to a second viewpoint on regional imbalances. The thinking behind the

cumulative–causation approach, and its reflection in the core–periphery model, has been criticized on several grounds [Gilbert and Gugler (1982)]. The role of foreign influences is underplayed, the poverty issue is neglected, the precolonial history of nations is ignored, and governments are assumed to operate "in the public interest". These lines of criticism from neo-Marxist writers [e.g. Stuckey (1975), Santos (1979)] argue for a recognition that the geography of development is a reflection of class interests, the extraction of surplus value over space, and the role of the state in supporting private capital. The driving force of the economy is capital accumulation and this imperative results in the co-existence of dominant core areas and subordinate peripheral regions. The dynamic national centre is in turn dependent upon the financial and industrial capitals of developed nations. The hierarchy is strengthened by the activities of social and political elites, making strong countervailing policies unlikely. Regional convergence has to wait for political and social development as well as for economic development.

Another normative model usually associated with radical views is the "agropolitan" view of regional development, also advocated by Friedmann [Friedmann and Douglass (1978)]. A key element is "selective regional closure", which implies a high degree of economic and political autonomy in a region, even at the expense, at least in the short run, of aggregate income growth [Stöhr and Tödtling (1978)]. This is combined with advocacy of the maximum possible decentralization of decision making. Agriculture and other indigenous natural resources are emphasized strongly in the development process.

2.4. *Neoclassical models*

A third viewpoint rests on a much narrower vision of the dominant forces in an economy. At its neoclassical simplest, this view takes regions as aggregated production units among which an equilibrium of income and employment emerges from the smooth working of factor markets. Imbalances merely reflect lags in the adjustment towards equilibrium (recognizing that the target may be moving faster than the arrow) or imperfections in market processes. In newly industrializing nations, it is easy to point to numerous reasons why one or both of these conditions may hold. A more developed neoclassical view allows for increasing returns and new technology, the influence of public-sector supporting investment, and a more dynamic perspective [e.g. Borts and Stein (1964), Romans (1965), Siebert (1969)]. It is then possible to develop this into a view of urbanization which rests on economies of agglomeration broadly defined, involving nodal rather than homogeneous regions [e.g. von Böventer (1970), Richardson (1973)].

This approach is the one with which the majority of regional economists feel most comfortable. However, critics point to the narrow explanation of regional

growth patterns, an excessive emphasis on the formal manufacturing sector as the prime agent of change, and the relative neglect of social and political factors. Also, systems of government and the complexities of state interventions in the economy found in all modern states have major locational consequences. These range from the physical location of the large public sector workforce to the hierarchy of power in the allocation of public expenditures and in the determination of tax rates and charges, to sites of access for permissions, permits, finance and protective subsidies, and to the spatial consequences of macro and sectoral policies.

3. Policy implications of the theories

The three approaches – cumulative causation, radical and neoclassical – are not mutually exclusive. Although they offer alternative theories of regional development, suggestive elements for regional economic development policies can emerge from all three. Approaches to policy design, however, tend to have a different emphasis from one theory to another. Crudely, the neoclassical view leans to incentives to steer productive factors, especially in the formal manufacturing sector. Cumulative causation theory leads to policies for growth centres and improvements in infrastructure, transport and communications, and education. The radical views suggest an emphasis upon regional closure and greater regional autonomy. A policy package can obviously contain elements of all these approaches, without a specific commitment to any of the theoretical under-pinnings.

3.1. Manufacturing bias

In the LDC context, however, two other aspects of economic development have been relatively neglected in discussions of regional policies. The first is the emphasis given in many national economic development strategies to the manufacturing sector, whether in import substitution or export promotion, in comparison with the resources devoted to the development of commercial agriculture. The message of Schultz (1964) and others, advocating higher prices and stronger policies to support innovations and farm-based investments generating large multiplier effects, has taken two decades to become the conventional wisdom in development economics [Little (1982), Tolley et al. (1982), Schultz (1978)]. The impact of active pro-agricultural policy measures has not been incorporated into regional development theory, although it has influenced the practice of Integrated Rural Regional Development planning.

3.2. *Primate city and core region bias*

The second area of neglect is very much related: it is that of the ("implicit") spatial impacts of non-spatial policies. Protectionist and subsidy policies for specific industrial sectors, infrastructure investment policies in response to need (rather than potential), policies for the uniform pricing of public services, and tax policy: all have locational effects. These effects are very difficult to identify, but where serious efforts have been made to quantify their impact [e.g. Tyler (1983)] the policies appear heavily biased in favour of the prosperous core region and the primate city.

3.3. *The dubious relevance of developed countries' experience*

An interesting question, which will be hotly debated [Chatterji (1982), Richardson (1980)], is the value of regional economics techniques and models for regional policy analysis for developing countries. Answers to this question raise several distinct issues. The first is whether the behavioural assumptions which underpin most regional models as applied to developed countries, particularly the modes of economic rationality that are most easily summarized as neoclassical assumptions, are appropriate to developing countries. Although tradition and culture may play an important role, for example in the agricultural and informal manufacturing sectors, there is increasing evidence that firms and households in developing countries respond to economic incentives in much the same way as in developed countries. On the other hand, the public sector is more interventionist in most developing countries (e.g. as a direct participant in production or as a price controller) which suggests that optimization models may be more useful than market models.

A second issue is whether developing countries have the data base and information systems to permit the application of sophisticated quantitative models. If not, new types of models using "soft" approaches such as applications of multidimensional scaling methods involving cross-impact matrices and other qualitative assessments [Nijkamp and Van Pelt (1983), Nijkamp (1983), Voogd (1981)] may be more relevant in developing country contexts. However, the range and quality of data varies widely among developing countries, from some African countries which have never had a population census to countries such as Brazil which have better regional and urban data than many developed countries. Nevertheless, in many developing countries subnational economic statistics are so rare or weak that innovative approaches in the use of qualitative information will be needed to evaluate regional policies, both ex ante and ex post.

A third problem is whether the assumptions about the stability of economic structures which underpin most regional economic models are relevant in devel-

oping countries where a primary objective of regional policy may be to transform the regional economy very rapidly indeed. Take, for example, interregional input–output models [Hewings (1983), Richardson (1985)]. The assumptions that interindustry and interregional trade coefficients remain constant or change in predictable ways may be invalid in developing countries, especially at the regional level. Of course, the operational flexibility of regional input–output models is so great that this limitation has not prevented their use in developing countries. Where data problems are compounded with inappropriate assumptions, however, the results can be unfortunate. A striking example is the 11-region input–output model developed in prerevolutionary Iran by the Battelle Institute. All the coefficients in this model were estimated using "Delphi methods". In practice, this involved asking informed people familiar with the individual sectors for their estimates of coefficient values. It is a little unkind but not inaccurate to say that all the coefficients in the model were invented!

4. Urban dimensions of regional policy

Regional problems in developing countries are an inevitable consequence of economic growth. If economic growth is dependent upon the links between a given country and the world economy, then regional economic differences within the nation are likely to be wider than in a closed economy. These economic forces also influence the pattern of urban development within the nation. The particular pattern of urban growth experienced may then influence in turn the relative economic fortunes of different regions. It is to this interdependence that we turn next.

In many countries strong interregional differences in the rate of expansion of towns and cities have emerged. This has led to concern with the dominance or primacy of one or a few major cities, and/or with pressures of expansion on individual cities which challenge the managerial abilities of urban authorities. Frequently, governments have attempted to influence the interurban distribution of this population growth. They have tried to slow down the growth of large cities, to decentralize growth, to promote the growth of secondary cities, to expand agricultural service centres or to generate industrial growth complexes. Increasingly, these policies are integrated under the banner of "National Urban Development Strategies" or "Population Distribution Policies" [Richardson (1981)]. These policies are complementary to regional economic development policies, each reinforcing the other. The two policy sets are analytically rather different, however.

Although regional development policy is the primary concern of this paper, we have not been able to avoid intermittent references to urban development. This

invasion of the territory of a companion volume requires some justification. This is not difficult. Almost every regional economist favours the concept of the nodal or polarized region to that of a homogeneous region, and nodal regions emphasize the linkages among urban centres both within and between regions. To the extent that regional development strategies focus on promoting manufacturing and quaternary services, these strategies must pay some attention to urban structure and facilities because most establishments in these sectors are located within urban centres. A key aspect of the growth centre concept is the relationship between the centre (or pole) and its hinterland [Richardson (1976)] which, in effect, makes the urban and regional dichotomy impossible. In geographically small countries interregional effects can barely be understood outside the context of the system of cities; even in very large countries ignoring this context is possible only in very special cases (e.g. sparsely populated frontier regions with very few urban centres such as the Amazon regions of Brazil or Peru). A final and compelling justification is that regional policy analysts may have to deal with urban–regional interactions by default, because urban economists often treat the city as an isolated production and consumption unit whose connections with the rest of the system (its home region, other cities and regions, the international economy) are not considered to be of interest except in the form of exogenous flows.

The spatial thrust of any regional economic development policy is seen in the relative growth rates of cities and urban areas, region by region. While urban growth is not always synonymous with economic growth, cities with rapidly growing populations usually experience a rapid growth of economic activity. If the spatial distribution of these growth rates is changing, both between regions and within regions, a major determinant of these changes is the growth of the formal manufacturing sector. In most regional development policies in the industrializing nations, priority is given to instruments which will influence the location of manufacturing. In only a minority of cases is there a complementary effort to promote development in the agricultural sector or the service sectors. There is a danger, therefore, in measuring regional development only in terms of new factories and the jobs they provide. However, investments in urban infrastructure and urban-based services af all kinds, which are required to promote manufacturing, also assist and induce development in other sectors. Urban growth is not just a multiplier consequence of manufacturing.

Pressures of population growth in nearly all developing countries not only present policy makers concerned with issues of spatial distribution with major operational problems (problems which may overwhelm informed and longer-term strategic choices); these pressures also provide an opportunity for influencing the geographical distribution of development. This opportunity rarely exists in the typical fully industrialized nation with a static population. United Nations estimates in 1980 [Hauser and Gardner (1980)] suggested that between 1980 and

the year 2000 there will be a 78 percent rise in the urban population worldwide, an increase of 1,400 million people, nearly three times the estimated rural increase. Three-quarters of this growth will occur in the less developed nations of the world, with an average rate of urban growth of 3.9 percent per annum. Although the growth rates in many countries have begun to decelerate, the absolute increments to urban populations remain very large. A major component of the "regional problem" in many developing countries is how to accommodate this expansion.

Urban growth, even when primarily fueled by natural increase rather than by net in-migration, presents opportunities for developing industry, raising productivity and incomes, and reducing poverty. Both internal and external economies of scale are exploited, innovations spread more rapidly, and the concentration of economic and political power [Friedmann (1972–73)] reinforces the advantages of the larger cities. However, there is widespread concern that the processes of urban growth, especially in the larger cities, are excessively rapid and costly. Rightly or wrongly, economic inefficiencies and inequities are seen as being increased by the continuing growth of large urban areas. Policies designed to improve regional "balance" would, on this argument, also serve an efficiency goal in seeking to redistribute urban growth. But most economic commentators are wary about supporting a regional development policy on these grounds. It is unjustifiable under strict economic efficiency criteria to advocate one urban size or one particular city size distribution across a region or a nation as being preferable to another.

In colonial and precolonial times, the cities of the low and middle income nations were centres of government, ports or agricultural trading centres. Industrial development has in many cases supplemented these historical roles, usually reinforcing the advantages of city size. This pattern of growth has had three consequences for regional development policies. One is a high degree of primacy, i.e. the concentration of urban and industrial growth in the single largest city (and hence in a single region). A second is a focus of urban growth in cities that are already large. A third is that a region may lag in urban growth because of its history rather than because of any inherent unsuitability for industrial development. In such cases, industrial development may be stimulated by improvements in transport and communications or by the opening up of agricultural and mineral resources to the market.

Against the background of these three urban growth consequences of industrial development, most advocates of regional development policies have insisted upon an urban growth strategy as a central component of a regional policy [e.g. Renaud (1981), Rondinelli (1983)]. This is in contrast to the industrial-development-focused policies of many developed countries. Different developing countries face contrasting situations in their levels of development and their geography, as the classification of Renaud (1981) makes clear. But all face a

choice of urban growth strategy. Richardson (1981) presents ten alternatives, from development axes to the promotion of secondary cities or smaller growth centres, recognizing that a hybrid strategy will most likely be appropriate for any given country. All strategies, except a laissez-faire approach, involve selectivity among cities for infrastructure allocations and development incentives. Indeed, the selection of too many centres as candidates for promotion, an understandable reaction to political pressures, has probably been the single most undermining factor in experiences with regional development policies to date. Competition between places and areas in claiming favoured treatment is a central feature of all regional policies, a conflict which perhaps diverts attention away from intra-area inequities and the levels of relative deprivation at the household level under which the goals of the policy are normally defined. Urban growth strategies can also have a built-in conflict. The most effective strategy for slowing the growth of a metropolitan core area in an economy is often the promotion of major secondary cities in the core region, i.e. intraregional decentralization. This at once conflicts with attempts to build up peripheral regions by diverting public sector urban-based expenditures and new industrial investments from the core region as a whole.

The linking of regional economic development to urban growth planning emphasizes the twin notions of layer and perspective in drafting a policy to respond to regional problems in developing countries. Friedmann (1975) has offered a list of issues of regional development planning:

(i) Regional coordination of national investment policies and programmes.

(ii) Excessive centralization in the territorial organization of power.

(iii) Growing urban primacy and underutilization of natural resources on the periphery of major core regions.

(iv) Underutilization of manpower and human talent.

(v) The incorporation of internally dominated peripheral regions into a single national economic system on a basis of greater equality of living levels and reciprocity in exchange.

(vi) Dualistic development of rural and urban economies.

(vii) Destructive use of land resources.

To these, we would add:

(viii) Efficient use of scarce skills in regional development and urban management.

(ix) Concern with the social effectiveness of urban infrastructure investment.

(x) The implicit spatial impact of non-spatial policy decisions.

This set of issues has to be addressed by national policies, but those policies have to be delivered at subnational levels (region, city, rural area and village). Regional economic development planning can be seen as offering a coordinating framework to associated policy areas at these various layers. In each policy area, however, a decision has to be reached about which functions and responsibilities

should be left to the control of each layer, and this is linked to decisions about levying taxes and allocating the revenues generated. It is here that perspective is important, not only for different views about the devolution of political power and the appropriate extent of administrative decentralization, but also in the diagnosis of the problems to be addressed. Much of the earlier writing in regional economics was non-spatial, in which regions were treated as discrete entities, in effect as competing economies. The influence of quantitative geography and regional science has since become important. The spatial structure of the national economy is now heavily stressed when defining regional problems, and the spatial structure of the region is taken into account in prescribing a policy response. In addition to these perspectives, more attention is being given to distributional issues, to be considered as equally important as economic efficiency and growth. The new visibility in the literature given to the internal management of cities in developing countries as an important determinant of the direction of urban growth makes a link to regional development analysis which has yet to be fully exploited [Linn (1983), Gilbert and Gugler (1982)].

5. Policy instruments and strategies

The instruments for promoting economic development in selected regions (and their major urban centres) include cash subsidies to capital and to labour, indirect protection from the tax system and import controls, the allocation of public sector investments, intervention in the location of public sector employment, licensing arrangements, the sponsorship of growth centres and new towns, and administrative decentralization. Direct controls over migration have been tried [and mostly failed; Simmons (1981)]. Granting a degree of political and fiscal autonomy to regions has been recognized as a means of giving an impetus to local economic development. Choosing an appropriate policy mix among the many options is not easy, although governments rarely experiment in regional policy innovation. This section of the chapter discusses these issues in the light of developing country experience in the last two decades.

5.1. Regional planning agencies

The need for a specialist development agency in a designated region will depend not only upon the formal structures of central, state and local governments but also upon their capacities. Relatively little appraisal of the performance of specialist agencies has been undertaken, taking into account the governmental context within which they work. The mere existence of an agency tends to give it an ex post justification which may be difficult to identify ex ante. Nevertheless,

the relative success stories in regional development planning may be linked to the creation of a multisectoral agency with special powers and responsibility. The examples of SUDENE in North East Brazil and SUDAM in the Amazon region are frequently quoted, though many local observers remain unconvinced that these particular agencies have worked well in practice.

Key elements in the success of specialized agencies are the size of their financial resources and the extent of their powers to decide the levels and scope of subsidies to industry and allocations for key infrastructure investments. Their role is most important in developing countries where governmental power and control is highly centralized. In such cases, little cross-referencing to other ministries takes place and the coordinated delivery of a carefully planned programme in a development region rapidly fragments. The need for coordination is one reason why regional development agencies are often not placed under a single central government or state government ministry but have been attached to the office of the prime minister or state governor.

In the typical developing country a regional economic development policy, whether or not it operates through a special agency, has two major coordination roles. The first is to coordinate planning for agricultural development and planning for industrial development, striving to exploit the potential linkages between the two sectors. The second is between sectoral planning and planning for urban growth. The difficult balance to be achieved in urban policy is between planning for the growth of services in response to excess demand and the need to take a strategic view of the allocation of urban investments to promote the growth in agricultural and industrial output (i.e. the trade-off between social infrastructure and economic infrastructure).

The coordination roles of special agencies in regional development can make their political control a difficult and contentious issue, with the clear temptation to place them under a form of executive board at arms length from politicians. This is particularly difficult when the agency is a central government creation, operating in both the geographical and the functional territory of a state or local government. The lower-tier authority may have been required to surrender powers to the agency. The potential for bureaucratic conflict is serious. The central government may have chosen to create the agency in response to a poor performance by lower-tier authorities as much as in response to lagging development or the identification of economic potential. Or the agency may also have been created in response to pressures from outside international lending agencies, such as the World Bank. An international institution, a foreign aid programme or an international commercial bank will try to minimize both its administrative costs and its risk. Not only is it simpler to deal with a single bureaucracy, but the financial accounting is more straightforward. The opportunities for the lender to press for policy reforms will also be greater than if the borrowed funds were channelled through several government departments.

5.2. The role of infrastructure

A major issue for regional policy in developing countries is the balance between economic infrastructure and social infrastructure in the allocation of scarce investment resources. One argument is that this should vary by region and over time. Hansen (1965) adopted a threefold classification of regions (congested, intermediate and lagging) and argued that infrastructure policy should focus on economic infrastructure in intermediate regions and on social infrastructure in lagging regions in the early phases of development. This view was supported by a case study of Mexico [Looney and Frederiksen (1981)]. First, public infrastructure investment on these lines appeared to stimulate GRP. Second, the emphasis on economic infrastructure in intermediate regions and on social infrastructure in lagging regions had a favourable effect on regional income disparities: thus, "the implicit trade-off between growth and minimizing differences in income need not exist" [Looney and Frederiksen (1981, p. 295)]. This last point is particularly important because it is well known that interregional disparities in social indicators (such as social service facilities, e.g. hospital beds, schools) are much wider in developing countries than disparities in per capita income. This provides a rationale for social infrastructure provision in peripheral areas. Side benefits might include: better prospects for the immigration of managers, professionals and technical workers into the periphery; and a reduction in the overall rate of periphery-to-core migration as a result of the narrowing of interregional *welfare* differentials. However, the linking of economic infrastructure investments to intermediate regions is consistent with arguments elsewhere in this chapter that infrastructure strategies have to be spatially selective and that regional policies should vary in focus from one kind of region to another. Finally, the overall scarcity of investment resources implies that the adoption of appropriate (usually lower) infrastructure standards and cost recovery mechanisms are as critical issues as the *type* of infrastructure to be emphasized. Reducing standards and recovering costs are important for both economic and social infrastructure (e.g. building access roads to industrial estates that are too wide is no more excusable than supplying water to households at prices that are too low).

5.3. Industrial promotion policies

It will be difficult for a regional development policy to help agricultural and industrial enterprises to be permanently viable unless initial financial injections and technical assistance are subsequently supported by an active commercial banking system. These banks could be in the public sector or the private sector, and they could be investment or development banks rather than consumer-oriented banks. The important issues are their accessibility for commercial and

industrial clients in the development region and the rules under which loans are given. The problems are more serious in rural regions without a large city. Many governments have recognized banking as a constraint on development in rural areas and small towns (e.g. the Nigerian Rural Banking Scheme, the Brazilian Rural Credit Outposts, and the Indian Regional Rural Banks).

The choice of policy instruments in a regional development programme has to be made within the contexts of existing urban and industrial policies, the level and pattern of economic development, and the government structure. The choice of instruments will also be influenced by the criteria used to measure their impact. Typical criteria include the generation of output or jobs, the leverage effect on private investment, the use of scarce budgetary resources, the degree of administrative complexity, the bias introduced into resource allocation, or anticipated local multiplier effects. There are marked differences on how different instruments score against these criteria [Townroe (1979, p. 118)].

Some hold the view that regional development in a country experiencing rapid economic expansion and population growth at the national level can be supply-led. This particularly North American sentiment was one of the reasons why such a large part of the regional development expenditures in Appalachia ended up in concrete for roads [Hansen (1972)]. It also explains why overinvestment in serviced industrial land has been so common in lagging regions in developed and developing countries alike. The policy conflict is clear. Infrastructure investments form necessary preconditions for all but the largest industrial investments to take place. The service or facility must exist before industrial operations can commence. But infrastructure investments typically exhibit strong economies of scale with large discontinuities in the cost function. Add the embarrassment of a failure to supply and the temptation to overinvest is clear. Infrastructure investments are also the controllable end of a regional development programme. The expenditures do not have to wait for a private sector response.

The most extreme case of supply-led infrastructure is the industrial estate. Industrial estates have been established in all parts of the developing world. In some countries, such as Thailand and Kenya, specialist agencies have been set up with the sole function of developing industrial estates. The concept of an industrial estate is basically sound. The spatial juxtaposition of industrial establishments, especially if they are from a similar set of industries or linked sectors, permits economies of agglomeration in the form of reductions in unit infrastructure costs and avoids higgledy-piggledy land use patterns. The problem is how to plan industrial estates in line with effective demand so that scarce investment resources are not tied up in underutilized infrastructure. The alternative approach – building a large estate from scratch and, after completion, sitting back and waiting for the firms to turn up – is very risky. The supply of industrial infrastructure may be a necessary condition for regional industrial development, but it is far from a sufficient condition.

Although many industrial estates in developing countries, especially in the primate city or its core region, are filled very quickly, there are many, many more which remain more or less empty for years. The main reasons for overbuilding are the poor quality of market analysis (overestimating the demand for new industrial space) and proliferation in the number of industrial estates in response to local political pressures. Too many industrial estates built too fast on sites which are too large but are often badly located (e.g. where cheap public land exists rather than at the optimal locations from the point of view of accessibility to interurban highways or workers), and using infrastructure standards that are too high or are otherwise inappropriate to needs: this is the story of industrial estate failures everywhere. The success rate could be improved dramatically by phasing, in terms of both the number of estates and the rate of construction of each estate. This is the only way of making an industrial estates programme compatible with economizing on scarce infrastructure resources. Also, it is unrealistic to expect even the more promising industrial estates in the peripheral regions of developing countries to be successful without supplementation with other measures, such as location subsidies.

5.4. Communications and human resource investments

Two important aspects of regional development which frequently fall outside the direct control of a regional development agency are communications and education. As a country starts to industrialize and both the road system and the telecommunications network are relatively underdeveloped, a lagging or peripheral region receives a measure of protection against industry and commerce in the core region from the friction of distance. Further investment in communications in and around the core region, on the other hand, increases its relative agglomeration advantages, developing secondary cities *within the region* as effective industrial locations competitive within the core city. Peripheral regions then suffer from their relatively restricted access. These regions require upgrading of their roads and other transport services and of their telephone networks if companies based there are to compete effectively for the core (or export) markets. But this upgrading simultaneously increases the exposure of the regional market to outside companies. Examples of the locational impact of major road developments and of the introduction of microwave telecommunication systems on industrial development can be found in many countries, e.g. Nepal [Blakie (1981)], or São Paulo State in Brazil [Dillinger and Hamer (1982)].

A familiar argument is that the task for regional policy makers will be made easier in the future as a result of technological advances, especially in the communications sector. This argument has three strands. First, technological progress diffuses over time through the national space economy, and this will give

the peripheral regions a chance to catch up (e.g. plants locating in the periphery will, on average, be newer and embody more recent technology). Second, a major obstacle to economic development in the periphery is the lack of economic infrastructure, such as transport and communications, and increasing wealth permits governments to invest in the periphery's infrastructure. Third, advances in communications are facilitating economic penetration of the periphery, by relaxing constraints on location. An implication is that even quaternary industries such as the information-processing industries will no longer have to rely on face-to-face contacts and hence to cluster in core region metropolises. They can become footloose and disperse to other regions, with relocation motivated by locational preferences or other reasons.

Whatever the merits of these arguments, it is doubtful whether they apply with much force to developing countries. A recent paper shows that, counterintuitively, as long as unit transport, communication and contact costs are higher in the periphery than in the core, the impact of technological progress in the information-processing sector will tend to widen core–periphery disparities rather than to narrow them [Clapp and Richardson (1984)]. This result is reinforced by the periphery's deficiencies in communications infrastructure, the locational pull on business services of the manufacturing sector which is heavily concentrated in the core region, the reluctance of professional and technical personnel to leave the primate city and its social and cultural amenities, and the inability of poor countries to pay international wage rates for computer programmers and similar skilled workers. Furthermore, adding capital in the form of computers and other electronic data-processing equipment will widen core–periphery income differentials by raising skilled wages in the core (reflecting capital-skilled labour complementarity) and depressing unskilled wages in the periphery (migration from the periphery to the core will slacken as a result of the substitution of capital for unskilled labour at the core).

The policy implications implied by this scenario are: heavy investment in interurban (interregional) telecommunications infrastructure; location incentives for business services and information-processing industries not only for manufacturing; and a national programme of human resource investments to increase the constrained supply of technicians in the business services sector. The underlying hypothesis is that eventually the high-order services sectors will become as important in the economic structures of developing countries as they are proving to be in the developed countries.

Manpower planning has had a chequered history in countries at all stages of development and it is difficult to advocate precise policies designed to generate skills in a workforce in advance of new industry expanding to take up the supply. At the same time, skill shortages have formed a major constraint on regional development initiatives, and many countries have felt a need to include an industrial training element in a regional development policy package. The local

multiplier effects of expenditures on training are high, unless the acquisition of a skill forms an incentive to migrate, an incentive which is obviously increased if there is no local opportunity to apply the skill. Hence the attraction of tying subsidies for skill training to on-the-job training or to company-specific off-the-job programmes rather than promting specific industrial skill training in colleges in advance of need.

All of the indirect and infrastructure elements of a regional development programme seek to attract external industrial investment and to foster local companies. Four questions have to be asked of each element: What is the appropriate scale of expenditure at each location? What level of quality is required for effectiveness? Can or should a price (or tax) be charged for the service? How important is consistency in delivery? The incentive effect of these expenditures on training, communications, utilities, and industrial parks is low but, to a greater or lesser degree, they form preconditions for development. It may not be necessary to match what is available in the core region, but it is necessary to provide enough infrastructure and services to allow industrial development to take place.

6. Recent regional development experiences in developing countries

Regional development incentives aim to attract mobile industrial investment from other regions or abroad as well as to promote the more rapid growth of indigenous firms. In some countries, incentives available in lagging regions have been combined with taxes or licensing procedures in the more prosperous areas, hoping to induce a flow of footloose plants away from the centre. The range of incentives available and their principal strengths and weaknesses have been reviewed in several studies of regional policy for developed countries [Armstrong and Taylor (1978), Richardson (1979), Emanuel (1973), Hansen (1974)]. Comment is limited here to issues which refer specifically to regional development in developing countries.

A serious practical problem in the design of locational incentives (and disincentives) is most easily understood in the context of the core–periphery model. In effect, the core–periphery model divides the national space economy into *two* regions (or two sets of region), namely the core and the periphery. A locational subsidy is intended to encourage industrial growth in the periphery (and, especially if it is combined with a locational tax, to discourage growth in the core region). But in geographical terms the two-region framework implies that where the core region ends the periphery begins. If the subsidy applies to all parts of the periphery, core region firms can obtain its benefits without bearing the higher risks and costs of a peripheral location by establishing plants a short distance across the core region boundary. This "boundary problem" distorts the impact of locational incentive schemes in many developing countries. Examples include:

the 50 kilometer ban on industrial development around Manila which has induced artificial industrial growth in secondary cities a few kilometers further out; the rapid buildup of industry at the Hub industrial estate where Karachi firms can obtain the subsidies offered to the backward province of Baluchistan; the growth of industries at Mirzapur in Bangladesh adjacent to Dhaka district which is excluded from industrial incentives; and many more. The solution to this problem, rarely adopted, is to create a buffer zone between the core region and the periphery in which no incentives are offered. However, to avoid discrimination against this buffer zone, disincentives (taxes) need to be imposed on industrial development in the primate city and the core region so that zero incentives within the buffer zone place it in an intermediate neutral position between encouraged industrial growth in the periphery and discouraged growth in the core.

The target group of firms for an incentive policy, even in the largest developing countries, is relatively small. The number of companies in the core region which might be tempted to decentralize or establish a branch plant unit in a peripheral region will be no more than a few score, even in cities the size of SãoPaulo, Bombay, Cairo or Bogota. The number of established companies within a designated region which might be encouraged to expand faster will be even smaller. The limited universe of manufacturing enterprises in the formal sector of these economies presents both a problem and an opportunity. The problem is the lack of choice and the scarcity of managerial skills in the companies able to respond to incentives and to the difficulties of multiplant operation. The opportunity lies in the limited number of contacts that have to be made and the possibility of tailoring incentives to the situation of individual companies. Both state-owned companies and multinational corporations bringing in investment from outside may receive special treatment. Many developed countries have found MNCs particularly receptive to regional incentives. This may be less true in developing countries. MNCs are typically at the high-technology and strong export ends of the industrial spectrum, and the relative agglomeration advantages (including access to government offices and agencies) of the core region outweigh any cost differentials resulting from subsidies to the periphery. For both multinationals and domestically owned concerns, the size of subsidies and the degree of protection available at the national level for new operations may be so large as to pose a severe visibility problem for even generous regionally based payments. This problem suggests that regional incentives should take a distinctively different form from any national incentives.

In developing countries government involvement with industry is strong, placing a premium on contacts with officials (and politicians). Extensive licensing is common for investment, imports, and the use of foreign exchange. For an expanding enterprise, the single most important constraint may be foreign exchange, introducing a new potential policy lever. A further feature of many

developing countries is a strong political commitment to economic growth, with the result that policy makers may be reluctant to introduce measures which inhibit economic growth and impair the efficient use of fiscal resources and capital. Unemployment or under-employment might be expected to make governments resist incentive policies which increase the relative capital intensity of projects, but administrative simplicity and a belief in relatively fixed factor proportions leads to the opposite conclusion. An intermediate response is to attach an employment creation, or cost-per-job, condition to capital subsidies.

Capital subsidies have been the prime instrument used to influence the location of manufacturing industry. One interesting variant is the points system used in North East Brazil by SUDENE. Under the 34/18 program [Goodman (1975)], capital subsidies have been available up to 75, 60, 50, 40 and 30 percent of total project investment costs, depending upon such factors as employment creation, the specific industrial sector, import substitution, export generation, local purchasing, location within the North East region, labour intensity, and the pattern of ownership. Similar ideas have been used in Malaysia [World Bank (1975)], linking regional development payments to national priority sectors, the Malaysian content of the product, and employment generation. A highly discretionary approach to incentives poses problems in the need for trained, skilled, knowledgeable and incorruptible civil servants to administer the scheme, and also violates Friedmann's (1972) requirement that the policy be simple, pragmatic and easy to understand. Experience with regional economic policies in both developed and developing countries has shown the need for a medium-to-long-term political commitment, avoiding frequent changes in either the form or intensity of the incentives payable.

One key finding of studies of industrial location decision making in developed nations is the importance of "non-economic" costs in locational choice. Large companies have to consider whether their top managers are willing to transfer to the new location. For the owner–manager, the upheaval of a household relocation may be too much to countenance on top of a plant relocation. Any policy oriented to industrial mobility in a newly industrializing country will face similar constraints. For example, in the 1980 São Paulo Industrial Location Survey [Townroe (1983)], "Personal and family reasons of the owner" were cited by 20 percent of firms decentralizing out of the São Paulo Metropolitan Area as a major or decisive factor in locational choice. This proportion rose to 29 percent for new firms, and to 42 percent for firms moving from locations in secondary cities in the State (the majority of this group consequently moved *within* the same secondary city). The factor "Favourable environment for attracting management staff" was not as important, however, being cited as a major or decisive factor by 19 percent of the decentralizing group, only 8 percent of the firms moving out to the suburbs but staying within the metropolitan area, and by 12 percent of the secondary city movers. The importance of these factors varies by size of firm, and

whether a branch plant or a complete transfer is being contemplated. However, in general, the "psychic distance" of a relocation from the primate city region of a developing country to a peripheral region is relatively greater than that facing a manager or owner in a highly integrated industrialized nation.

A United Nations working group on policy instruments for decentralized industrialization [UNCRD (1976)] suggested satisfactory results could only be obtained with direct subsidies, strong fiscal incentives, the development of financial intermediaries, pushing of foreign investors to less developed areas and the pricing of public utilities to support decentralization. This conclusion, however, appears excessively influenced by the experience of West European regional development policies of the 1960s and early 1970s, failing to recognize the severe constraints of limited public sector resources in LDCs on the feasibility of alternative regional policy proposals.

Advocates of selective regional closure as the key to development and improved regional balance, in the attempt to swing the pendulum more towards bottom-up approaches rather than top-down policy initiatives, run the risk of overturning economic and institutional relationships in an economy in order to satisfy limited regional development objectives. However, policy elements implied by this perspective, which emphasises "territory" rather than "function", extend the range of policy instruments for the promotion of regional economic development.

For example:

"(1) Provision of broad access to land and other territorially available natural resources as the key production factors in most less developed areas....

(2) The introduction of new, or revival of the old, territorially organized structures for equitable communal decision-making on the integrated allocation of regional natural and human resources....

(3) Granting a higher degree of self-determination to rural and other peripheral areas....

(4) Choice of regionally adequate technology oriented towards minimizing waste of scarce, and maximizing use of regionally abundant, resources....

(5) Assignment of priority to projects which serve the satisfaction of basic needs of the population....

(6) The introduction of national pricing policies which offer terms of trade more favorable to agricultural and other typically peripheral products....

(7) ...external assistance (national or international) would be solicited but should be mainly considered as compensation for the eroding effects of previously emerging dependencies....

(8) Development of productive activities exceeding regional demand... should be promoted only to the extent that they lead to a broad increase in living levels of the population of the territorial unit....
(9) Restructuring of urban and transport systems to improve and equalize access of the population to all parts of the country to them, rather than strengthening such systems oriented to the outside....
(10) Improvements of rural-to-rural and rural-to-village transport and communication facilities (rather than the present priority given to rural-to-large-urban communities) should have preference....
(11) Egalitarian societal structures and a collective consciousness are important prerequisites for a strategy for development "from below"." [Stöhr (1981, p. 64), see also Stöhr and Tödtling (1977, 1978), Lo and Salih (1981), Friedmann and Weaver (1979)].

It is difficult to envisage these policies applied to one or two selected regions and not to all. The discussion of bottom-up strategies therefore shades into arguments about appropriate national government structures. It is also not clear from the arguments put forward by this school of thought how decentralization and greater regional autonomy would reduce interregional equity as opposed to making existing spatial inequities more tolerable. Also, the cost in terms of national economic growth may not be acceptable to a national government. Selective spatial closure theorists have broadened the debate on the content of a regional development program, but the issues raised are more ethical and political than economic [Weaver (1981, p. 93)].

Without broadening the regional policy issue to national dimensions, there are four other policy avenues via which faster growth in a lagging region might be promoted. The first is a set of instruments directed at the urban and industrial complements to agricultural development, fostering services and agro-related urban-based manufacturing to accelerate the growth of agricultural output. The second is the use of public sector manufacturing industry as propulsive plants or sectors in designated growth areas, with the possible risk of underwriting locations that may be inefficient *intra*sectorally. The third avenue lies in the promotion of entrepreneurship in small-scale industry. Experience suggests that small-firm programmes are best delivered at the regional rather than the national level. These programmes require a rather delicate mix of advice, finance and credit, the provision of work spaces, and ongoing relationships between the small firms and the sponsoring agency. The last avenue is the decentralization of public sector employment, overcoming the objections of employees and relocating hospitals, universities, and government offices involved in routine clerical work to peripheral regions. There are probably communications diseconomies involved, but these activities are large employers with strong local multiplier effects. A "decentralization of government establishments" strategy is much more spatially

selective than administrative decentralization in general which involves the transfer of functions and personnel to *all* subnational government units. The latter's benefits are more political rather than economic. The choice between the two depends upon institutional constraints and policy objectives. Either is much preferable to a third type of government locational change, namely relocation of the national capital. Experience in Brazil and Pakistan suggests that this is a very expensive way of achieving limited regional development objectives.

7. Conclusions

In developed and developing countries alike, the world-wide recession of the early 1980s illustrated the difficulty of running an effective regional economic development policy in an environment of slow national economic growth [Richardson and Turek (1984)]. Most regional policies rely heavily on new industrial investments as a precondition for success.

However, in developing countries economic recession did not have a significant impact on the scale or pattern of urbanization. The underlying dynamic remains the fertility and mortality experience of the existing urban population rather than a slowdown in the growth rate of the GNP. Data are not yet available to test whether the deceleration in the growth of the large-scale manufacturing sector modified the migration streams from the rural areas and the small towns to big cities and from peripheral regions to core regions.

One lesson from the past two decades however is that the economic fortunes of a lagging region, and the potential for a policy intervention, are governed by external forces and institutions as much as by internal structures. One important external force has been a changing policy stance towards agriculture. Food production is now more highly regarded as a major contributor to economic development rather than being treated merely as a source of cheap sustenance, tax revenue and labour for urban manufacturing. The social benefits from land reform and higher agricultural prices are maximized if supported by investment in commercial inputs and by spending on rural services [electricity, water supply, transport, health, education, and technical assistance; see Johnson and Kilby, (1975) and Little (1982)]. This has implications for the urban development of rural regions [Rondinelli and Ruddle (1978)]. In the rural areas of Africa and Asia "Integrated Rural Regional Development" is interpreted increasingly in these agricultural support terms [e.g. Livingstone (1979), Hazlewood and Livingstone (1982), Belshaw (1977)].

The recent emphasis on rural planning reflects the more general problem of the difficult balance in regional development policies in developing countries among physical planning, public sector economic planning, and stimulus to an often unpredictable private sector. Policy instruments such as the provision of in-

frastructure or locational incentives cannot dominate or counter the commercial requirement of access to markets or to resources. A regional economic policy cannot be deterministic, and regional "blueprints" for development are an inappropriate conception of effective strategy. Regional development planning needs to be comprehensive in its coverage without the pretence of being comprehensive in what can be directed or coordinated.

Changes in thinking about the nature of regional economic planning and about the effective promotion of rural development are reflected in the cyclical attitude to the growth pole concept. Hansen (1981, p.34) suggests three phases: "(1) Optimism with respect to possibilities for inducing growth in a few centres and to the subsequent generation of spread effects; (2) pessimism when the expectations of the early phase failed to materialize; and (3) a broader view of growth centres as one aspect of more comprehensive development planning." Richardson (1978) also returned to a defence of the growth centre approach in the context of both rural development and national urban policy.

The growth pole approach has to be treated very cautiously in developing countries. There are many unsatisfactory experiences resulting from a careless transfer of growth pole policy ideas from developed countries with little adjustment to indigenous conditions. The most extreme cases have been based on heavy industrial complexes with negligible linkages with the hinterland and minimal employment generation. An alternative strategy might begin with the fact that most growth poles in developing countries are located in rural regions. Such poles might emphasize the growth of agro-based industries to expand the market for hinterland output, and hence to stimulate rural productivity and income, and the generation of higher-order services [Richardson (1978)]. This is not quite the same as the agropolitan development planning approach, because it implies intermediate-size centres somewhere between the large industrial poles implied by traditional growth pole theory and the small agropolitan centre.

Some aspects of growth pole theory are especially applicable in developing countries. The development axis concept [Pottier (1963)], for example, where the promotion of multiple growth poles (initially at the endpoints of the axis, but subsequently at intermediate distances along the axis) helps to generate the cost reductions and agglomeration economies associated with opening up a transportation corridor, may provide a successful means of integrating growth pole strategies and interregional transport investments. Similarly, countermagnet strategies work better in situations where cities grow rapidly via in-migration from rural areas than where people are reshuffled within an established urban hierarchy in a highly urbanized society. The idea of strengthening a regional metropolis as a counterweight to the primate city is more effective if the city size distribution is still evolving than in countries where the urban hierarchy has become fossilized. Unfortunately, the primate economies where countermagnets are most needed (e.g. Thailand, Peru, Chile, Argentina) lack the large candidate

cities for countermagnets, while those countries which have strong second and third cities (e.g. India, Malaysia, the Philippines) have little reason to adopt a countermagnet strategy. Also, a successful countermagnet may replicate the adverse spatial impacts of primacy in another part of the country.

The evolutionary character of national urban hierarchies in developing countries has another major implication for the implementation of growth pole strategies. Too often, growth poles have been selected in a piecemeal manner as "points" of development with little regard for each pole's role in the cementing of urban hierarchies, both national and regional. Growth pole policies cannot be implemented successfully outside the broader context of a national urban development strategy. Agglomeration economies are urban in character [both intra-urban and interurban; von Böventer (1975)], and the promotion of national urban development strategy will have direct repercussions on regional development, both within and among regions. Growth pole strategies, whatever their weaknesses, help to integrate urban and regional development.

After an earlier preoccupation with the role of metropolitan centres and small towns and villages in regional development, recent attention has focused on policies to improve the development prospects of medium-sized (secondary, intermediate) cities. This attention reflects the desire of national governments to diffuse urbanization spatially, but it also reflects a realization that secondary city policies can be a mechanism to reduce regional inequities as well as to improve administrative efficiency, raise urban productivity and reduce urban poverty [Rondinelli (1983)]. The scope of these programs (e.g. Brazil, Thailand, Indonesia) has included the improvement of local government managerial capacity, greater local financial autonomy, physical infrastructure investments, and the promotion of the economic base of each city. The economic dynamics of the growth of a system of secondary cities with different economic structures and roles and their regional implications is poorly understood. A recent empirical study of Brazil is an attempt to locate the sources and interactions of growth in these cities in rapidly industrializing regions which are larger than in most countries [Henderson (1983)].

As suggested above, many recent commentators on regional economic policy in developing countries have stressed non-economic institutional and political aspects of development planning, especially in rural areas. This has been a reaction to the apparent failure of Western urban–industrial models of regional development to reduce regional inequalities and rural poverty. This reaction has stressed several layers of dualism, between rural areas and urban areas in many developing countries, between the formal and informal sectors of the economy and between the industrialized and industrializing countries [Lo and Salih (1981)]. One benefit has been a greater understanding of the significance of political structures and the relevance of alternative administrative modes [e.g. Hinchcliffe (1980) on Papua–New Guinea]. A more dubious result has been

advocacy and polemics [e.g. Weaver (1981), Friedmann and Douglass (1978)], especially because of a refusal to acknowledge the role of innovation diffusion and the gains from trade, while playing down efficiency considerations in favour of redistribution and political devolution.

This paper began with a standard warning about the dangers of generalization. The regional problems of countries with large industrial sectors, whether middle-income nations such as Brazil or Mexico or poorer nations like India or China, will be rather different from the regional problems of nations with relatively small industrial sectors, such as Bangladesh or the Ivory Coast. However, the common problems of large agricultural sectors in the economies of all developing countries and of more or less rapid urbanization allow many perspectives to be transferred with some qualifications from country to country. The dominant task of regional economists in the medium-term future continues to be to build bridges between research in development economics and new insights in regional analysis and the economics of urban growth. The literature in this area is thin on long-term empirical studies of individual developing countries. The stage of merely applying techniques and policy findings generated in the developed countries is over.

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